Unit 2 Targeted Practice — Scientific Notation — Answer Key

Complete independently · Scientific Notation

A. Scientific Notation — Conversions

Problem 1 Score 3

Write 81,000 in scientific notation.

Answer: 8.1×10^{4}

Problem 2 Score 3

Write 0.00056 in scientific notation.

Answer: 5.6×10^{-4}

Problem 3 Score 3

Write 9.4×10^{5} in standard form.

Answer: 940,000

Problem 4 Score 4

Write 6.02×10^{-3} in standard form.

Answer: 0.00602

Problem 5 Score 4

Write 12,300,000 in scientific notation.

Answer: 1.23×10^{7}

B. Scientific Notation — Operations

Problem 1 Score 4

Multiply: $(4 \times 10^2) \times (2 \times 10^6)$

Answer: 8×10^8

Problem 2 Score 4

Divide: $(9 \times 10^8) \div (3 \times 10^2)$

Answer: 3×10^6

Problem 3 Score 5

Multiply: $(5 \times 10^3) \times (6 \times 10^4)$ — adjust if needed!

Answer: 3×10^8

C. Word Problem

Problem 1 Score 5

Light travels 1.86×10^5 miles per second. How far does it travel in 10 seconds? (scientific notation)

Answer: 1.86×10^6

Problem 2 Score 5

A farm has 2.5×10^4 trees. Each tree grows about 4×10^2 apples. How many apples in all? (scientific notation)

Answer: 1×10^7

Problem 3 Score 5

Sound travels 3.4×10^2 meters each second. There are 3.6×10^3 seconds in an hour. How many meters does sound travel in 3 hours?

A. 3.672×10^5

B. 36.72×10^5

C. 3.672×10^6

D. 1.224×10^6

Answer: C

Problem 4 Score 5

A stadium holds 6×10^4 fans. Each ticket costs 5×10^1 dollars. Ava says a sold-out game brings in 3×10^5 dollars. Is that reasonable?

A. Yes. $6 \times 5 = 30$, so it starts with 3, and the exponent is $4 + 1 = 5$.

B. No. It is 3×10^4 dollars, because $4 \times 1 = 4$.

C. Yes. 60,000 fans times $50 is about $300,000.

D. No. $6 \times 5 = 30$ and $10^4 \times 10^1 = 10^5$, so it is $30 \times 10^5 = 3 \times 10^6$ dollars.

Answer: D